

BAULKHAM HILLS HIGH SCHOOL

MARKING COVER SHEET



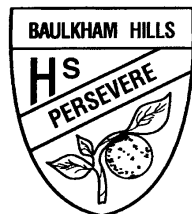
YEAR 12 HALF-YEARLY EXTENSION 1 2011

STUDENT NUMBER: _____

TEACHER NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
TOTAL	/ 84
PERCENTAGE	%

BAULKHAM HILLS HIGH SCHOOL



YEAR 12 HALF-YEARLY MATHEMATICS EXTENSION 1 2011

General Instructions

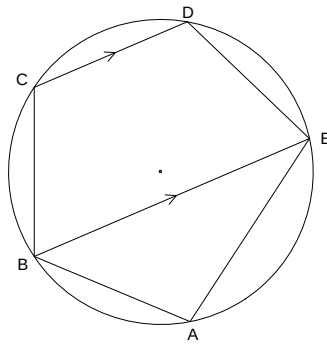
- Exam time – 2 hours
- Reading time – 5 minutes
- Start each question on a new page
- All necessary working should be shown
- Write your student number at the top of each page of your answers
- Board approved calculators may be used
- Write using black or blue pen

Total Marks: 84

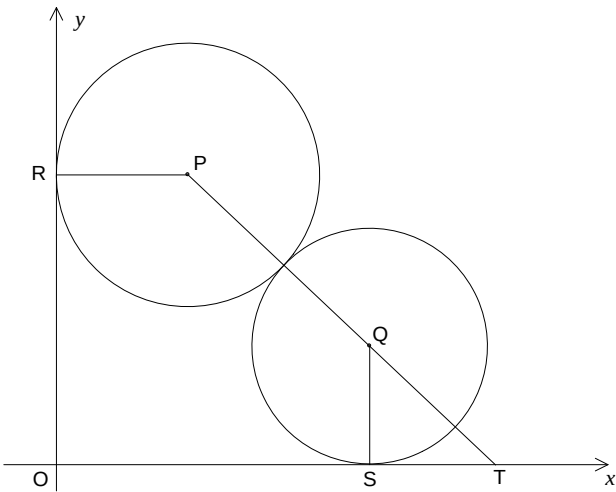
Attempt ALL questions

Question 1 Start on a new page - (12 marks)		Marks
a)	$\int_0^{1.5} \frac{dx}{\sqrt{9-2x^2}}$ leaving your answer in exact form	3
b)	Solve $\sin 2x = \tan x$ for $0 \leq x \leq \pi$	3
c)	Solve $x^2 + x + \frac{12}{x^2 + x} = 8$	3
d)	i) Find the derivative of $f(x) = \tan(x^2)$	1
	ii) Hence or otherwise evaluate $\int_0^1 x \sec^2(x^2) dx$ correct to two decimal places.	2
Question 2 Start on a new page - (12 marks)		
a)	Show that $\frac{d}{dx} \sin^{-1}(\sqrt{x}) = \frac{1}{\sqrt{4x-4x^2}}$	3
b)	Find the size of the acute angle, in radians, between the two curves: $y = x^2$ and $y = 8 + \frac{x^2}{2}$ at their point of intersection in the first quadrant. Give your answer correct to two decimal places.	4
c)	For the function $f(x) = 2 \cos^{-1} 3x$	
	i) Write down the domain and range	2
	ii) Draw a neat sketch showing all important features	1
	iii) Calculate the area enclosed by $y = 2 \cos^{-1} 3x$, the line $x = 0$ and the line $y = 0$	2
Question 3 (12 marks) - Start a new page		
a)	If $y = x^n e^{ax}$ where a, n are constants	
	i) find $\frac{dy}{dx}$	1
	ii) show that $\frac{dy}{dx} - ay = \frac{ny}{x}$	2
b)	Find k if $\int_0^{\frac{\pi}{6}} \frac{\cos x}{1 + \sin x} dx = \log_e k$	3
c)	Without the aid of a calculator find the exact value of $\tan\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$	2
d)	The arc of the curve $y = \cos 3x$ between the lines $x = 0$ and $x = \frac{\pi}{6}$ is rotated about the x -axis. Find the volume of the solid formed.	4

Question 4 (12 marks) - Start a new page		Marks
a)	Joel puts \$500 into a Baulko Bank for 2 years where it earned interest at 6%pa paid twice a year. He withdrew all his money and immediately deposited it into Axel Credit Union where his money earned 8%pa paid quarterly. If he withdrew his savings and had \$633.75, how long was the money kept in the credit union?	4
b)	The rate at which a body warms in air is proportional to the difference between the temperature, T, and the constant temperature, T_0, of the surrounding air. This rate is represented by $\frac{dT}{dt} = k(T - T_0)$ where t is the time in minutes and K is a constant. i) Show that $T = T_0 + Ae^{kt}$ is a solution of $\frac{dT}{dt} = k(T - T_0)$ where A is a constant. ii) For a particular body, when $t=0$, $T= 5$ and when $t= 20$, $T= 15$. Given $T_0 = 25$, find the temperature after a further 30 minutes have elapsed. Give your answer to the nearest degree. iii) Briefly describe the behaviour of T as t becomes large.	1 3 1
c)	Use mathematical induction to prove that for all integers $n \geq 1$ $\cos(x + n\pi) = (-1)^n \cos x$	3
Question 5 (12 marks) - Start a new page		
a)	The following eight tiles are taken from a scrabble set: A, A, B, B, C, D, E, F. How many different 4 letter permutations can be formed from these eight tiles?	3
b)	Find $\lim_{x \rightarrow 0} \frac{\sin \pi x^\circ}{x}$	3
c)	Find the following indefinite integral $\int \frac{x^2 + 1}{x - 1} dx$	2
d)	Sketch the polynomial function $y = P(x)$ in the domain $-3 \leq x \leq 3$ given that: $y = 0$ only when $x = 0$ and 2 $y' = 0$ only when $x = \pm 1$ $y'' = 0$ only when $x = -1$ and 0 $y'' < 0$ only when $-1 < x < 0$	4

Question 6 (12 marks) - Start a new page	Marks
a)  A, B, C, D, E are points on the circumference of a circle such that $CD \parallel BE$. Prove that $\angle CAB = \angle DAE$.	4
b) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ i) Show that the equation of the normal to the curve of the parabola at the point P is $x + py = 2ap + ap^3$ ii) Find the coordinates of the point Q where the normal at P meets the y-axis iii) Find the coordinates of the point R which divides PQ externally in the ratio 2: 1 iv) Find the Cartesian equation of the locus of R and describe the locus in geometrical terms	2 1 2 3

QUESTION 7 ON NEXT PAGE

Question 7 (12 marks) - Start a new page	Marks
a) Given that $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(1 - x)$ have values between 0 and $\frac{\pi}{2}$ i) Show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$ ii) hence or otherwise solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1 - x)$	2 2
b) The diagram shows two touching circles with centres P and Q. The circle with centre P has a radius of 4 units and touches the y-axis at R. The circle with centre Q has a radius of 3 units and touches the x-axis at S. PQ produced meets the x-axis at T and $\angle QTS = \theta$  i) Show that $OR = 3 + 7 \sin \theta$ and $OS = 4 + 7 \cos \theta$ ii) Show that $RS^2 = 42 \sin \theta + 56 \cos \theta + 74$ iii) Hence express RS^2 in the form $74 + r \cos(\theta - \alpha)$ Clearly stating the values of r and α iv) Find the maximum lengths of RS and the value of θ for which this occurs.	2 2 3 1

End of Paper

$$\begin{aligned}
 a) &= \int_0^{1.5} \frac{1}{\sqrt{x}} \frac{dx}{\sqrt{9-x^2}} \\
 &= \frac{1}{\sqrt{2}} \left[\sin^{-1} \left(\frac{\sqrt{2}x}{3} \right) \right]_0^{1.5} \\
 &= \frac{1}{\sqrt{2}} \left(\sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0 \right) \\
 &= \frac{\pi}{4\sqrt{2}} \text{ or } \frac{\pi\sqrt{2}}{8}
 \end{aligned}$$

$$\begin{aligned}
 b) \quad 2 \sin x \cos x &= \frac{\sin x}{\cos x} \\
 2 \sin x \cos^2 x &= \sin x \\
 \sin x (2 \cos^2 x - 1) &= 0 \\
 \therefore \sin x = 0 \quad 2 \cos^2 x = 1 \\
 \sin x = 0 \quad \cos^2 x = \frac{1}{2} \\
 \cos x &= \pm \frac{1}{\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore x = 0, \pi \quad \text{or} \quad x = \frac{\pi}{4}, \frac{3\pi}{4} \\
 \therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi
 \end{aligned}$$

$$\begin{aligned}
 c) \text{ let } m &= x^2 + x \quad \text{NB } x^2 + x \neq 0 \\
 \therefore m + \frac{12}{m} &= 8 \quad \text{ie } x(x+1) \neq 0 \\
 &\quad \therefore x \neq 0 \text{ or } -1 \\
 m^2 - 8m + 12 &= 0 \\
 (m-6)(m-2) &= 0 \\
 m = 2 \quad \text{or} \quad m = 6 \\
 x^2 + x - 2 = 0 \quad x^2 + x - 6 = 0 \\
 (x+2)(x-1) = 0 \quad (x+3)(x-2) = 0 \\
 x = 1, -2 \quad x = -3, 2 \\
 \therefore x = -3, -2, 1, 2
 \end{aligned}$$

$$d) i) f(x) = 2x \sec^2(x^2) \quad \checkmark$$

$$\begin{aligned}
 ii) &\frac{1}{2} \int_0^1 2x \sec^2(x^2) dx \\
 &= \frac{1}{2} \left[\tan(x^2) \right]_0^1 \\
 &= \frac{1}{2} (\tan 1 - \tan 0) \\
 &= 0.78 \text{ (2dp)}
 \end{aligned}$$

$$Q2 a) \frac{d}{dx} (\sin^{-1}(x^2))$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{1-(x^2)^2}} \times \frac{1}{2} x^{-1/2} \\
 &= \frac{1}{2\sqrt{x}\sqrt{1-x}} \\
 &= \frac{1}{\sqrt{4x(1-x)}} \\
 &= \frac{1}{\sqrt{4x-4x^2}}
 \end{aligned}$$

$$\begin{aligned}
 b) \quad x^2 &= 8 + \frac{x^2}{2} \\
 2x^2 &= 16 + x^2 \\
 x^2 &= 16 \\
 x &= \pm 4
 \end{aligned}$$

$$\begin{aligned}
 \frac{d}{dx}(x^2) &= 2x \quad \frac{d}{dx}\left(8 + \frac{x^2}{2}\right) = x \\
 m_1 &= 8 \quad m_2 = 4 \\
 \therefore \text{angle } \tan \theta &= \left| \frac{8-4}{1+8 \times 4} \right| \\
 \tan \theta &= \frac{4}{33} \\
 \theta &= \tan^{-1} \frac{4}{33} \\
 &= 0.12 \text{ radians}
 \end{aligned}$$

$$2c) i) \frac{y}{2} = \cos^{-1} \{x\}$$

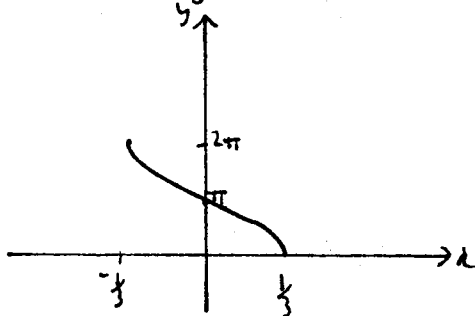
$$\text{Domain: } -1 \leq x \leq 1$$

$$\text{Domain: } -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$\text{Range: } 0 \leq \frac{y}{2} \leq \pi$$

$$\text{i.e. } 0 \leq y \leq 2\pi$$

(ii)



$$iii) y = 2 \cos^{-1} \{x\}$$

$$\frac{y}{2} = \cos^{-1} \{x\}$$

$$2x = \cos \frac{y}{2}$$

$$\text{Area} = \int_0^{\pi} \frac{1}{3} \cos \frac{y}{2} dy$$

$$= \frac{1}{3} \left[2 \sin \frac{y}{2} \right]_0^{\pi}$$

$$= \frac{2}{3} (\sin \frac{\pi}{2} - \sin 0)$$

$$= \frac{2}{3} (1 - 0)$$

$$\therefore \text{Area} = \frac{2}{3} \text{ unit}^2$$

$$3a) i) \frac{dy}{dx} = e^{ax} \cdot nx^{n-1} + x^n \cdot ae^{ax}$$

$$= x^{n-1} e^{ax} (n + ax)$$

$$ii) \frac{dy}{dx} - ay$$

$$= x^{n-1} e^{ax} (n + ax) - ax e^{ax} x^n$$

$$= x^{n-1} e^{ax} (n + ax - ax)$$

$$= n x^{n-1} e^{ax}$$

$$= \frac{ny}{x} \checkmark$$

$$\therefore \frac{dy}{dx} - ay = \frac{ny}{x}$$

$$3b) \int_0^{\pi/6} \frac{\cos x}{1 + \sin x} dx$$

$$= \left[\log(1 + \sin x) \right]_0^{\pi/6}$$

$$= \log(1 + \sin \frac{\pi}{6}) - \log(1 + \sin 0)$$

$$= \log(1 + \frac{1}{2}) - \log 1$$

$$= \log(1.5)$$

$$\therefore k = 1.5$$

$$3c) \tan(\cos^{-1}(-\frac{\sqrt{3}}{2}))$$

$$= \tan(\pi - \frac{\pi}{6})$$

$$= -\tan(\frac{\pi}{6})$$

$$= -\frac{1}{\sqrt{3}} \text{ or } -\frac{\sqrt{3}}{3}$$

$$3d) v = \pi \int_0^{\pi/6} y^2 dx$$

$$= \pi \int_0^{\pi/6} \cos^2 3x dx$$

$$= \pi \int_0^{\pi/6} \frac{\cos 6x + 1}{2} dx$$

$$= \frac{\pi}{2} \left[\frac{\sin 6x}{6} + x \right]_0^{\pi/6}$$

$$= \frac{\pi}{12} \left[\sin 6x + 6x \right]_0^{\pi/6}$$

$$= \frac{\pi}{12} \left[(\sin \pi + \pi) - (\sin 0 + 0) \right]$$

$$= \frac{\pi^2}{12}$$

4a) let A_n be amount after n 6 months

$r = 3\%$ per period

$$A_4 = 500(1.03)^4$$

$$A_4 = \$562.75$$

$$633.75 = 562.75(1.02)^n$$

$$\frac{633.75}{562.75} = (1.02)^n$$

$$\ln\left(\frac{633.75}{562.75}\right) = \ln(1.02)^n$$

$$n = \frac{\ln\left(\frac{633.75}{562.75}\right)}{\ln(1.02)}$$

$$n \approx 6$$

$\therefore$ money is in credit union for 1.5 yrs.

$$4(b) i) \frac{dT}{dt} = kAe^{kt}$$

$$= k(T_0 + Ae^{kt} - T_0)$$

$$= k(T - T_0)$$

ii) when $t=0, T_0 = 5^\circ\text{C}$

$$T = 25 + Ae^{kt}$$

$$5 = 25 + Ae^0$$

$$A = -20$$

$$\therefore T = 25 - 20e^{kt}$$

when $t=20, T=15$

$$15 = 25 - 20e^{20k}$$

$$20e^{20k} = 10$$

$$e^{20k} = \frac{1}{2}$$

$$\ln e^{20k} = \ln\left(\frac{1}{2}\right)$$

$$20k = \ln \frac{1}{2}$$

$$k = \ln(0.5) / 20$$

When $t=50$

$$T = 25 - 20e^{50k}$$

$$= 21.46...$$

$$T = 21^\circ\text{C}$$

(iii) As $t \rightarrow \infty$ $T = 25 - 20e^{kt}$

approaches $T = 25 - 0$ since $e^{kt} \rightarrow 0$

$$T = 25^\circ\text{C}$$

$\therefore$ Temperature approaches 25°C which is the temperature of the surrounding air.

c) Test $n=1$

$$\text{LHS} = \cos(x + \pi)$$

$$= \cos x \cos \pi - \sin x \sin \pi$$

$$= -\cos x$$

$$\text{RHS} = (-1)^1 \cos x$$

$$= -\cos x$$

$$= \text{LHS}$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\text{ie } \cos(x + k\pi) = (-1)^k \cos x$$

We wish to prove $\cos(x + (k+1)\pi) = (-1)^{k+1} \cos x$

For $n=k+1$

$$\cos(x + (k+1)\pi)$$

$$= \cos(x + k\pi + \pi)$$

$$= \cos(x + k\pi) \cos \pi - \sin(x + k\pi) \sin \pi$$

$$= -\cos(x + k\pi) - 0$$

$$= -(-1)^k \cos x$$

$$= (-1)^{k+1} \cos x \text{ as req'd.}$$

$\therefore$ If true for $n=k$ its true for $n=k+1$. But true for $n=1$, therefore true for $n=1+1=2$ and $n=2+1=3$ and so on for $n=1, 2, 3, \dots$

5a) Case 1 all letters different

A, B, C, D, E, F

$${}^6P_4 = 360$$

Case 2 Only 2 the same, 2 different

$$2 \times {}^5C_2 \times \frac{4!}{2!} = 240$$

Case 3 Both As and Bs

$$\frac{4!}{2!2!} = 6$$

$$\therefore \text{number of ways} = 360 + 240 + 6 = 606$$

$$5b) \lim_{x \rightarrow 0} \sin \frac{\pi x}{x}$$

$$= \lim_{x \rightarrow 0} \sin \frac{\pi \left(\frac{\pi x}{180} \right)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \left(\frac{\pi^2 x}{180} \right)}{\frac{\pi^2 x}{180}} \times \frac{\pi^2}{180}$$

$$= \frac{\pi^2}{180} \lim_{x \rightarrow 0} \frac{\sin \frac{\pi^2 x}{180}}{\frac{\pi^2 x}{180}}$$

$$= \frac{\pi^2}{180} \times 1$$

$$= \frac{\pi^2}{180}$$

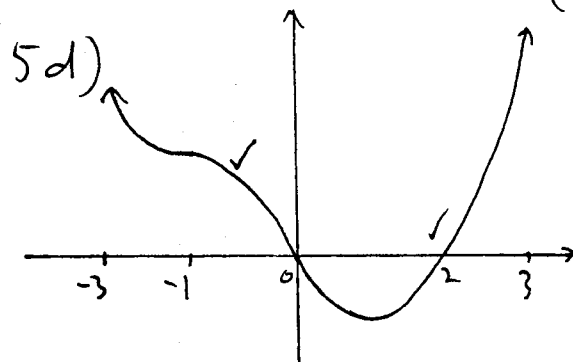
$$5c) \int \frac{x^2+1}{x-1} dx$$

$$= \int \frac{x^2-1+2}{x-1} dx$$

$$= \int \frac{x^2-1}{x-1} + \frac{2}{x-1} dx$$

$$= \int (x+1) + \frac{2}{x-1} dx$$

$$= \frac{x^2}{2} + x + 2 \log_e(x-1) + C$$



(1) x intercepts

(1) concave down between -1 and 0

(1) stationary points

(1) concave up $-3 \leq x < -1$ and $x > 0$

6a) Join AC, AD

$$\text{Let } \angle OLB = x^\circ$$

$$\angle CBE = 180^\circ - x^\circ \text{ (co-interior angles, } CD \parallel BE)$$

$$\angle CDE = x^\circ \text{ (opposite angles in cyclic quad } COEB \text{ are supplementary)}$$

$$\angle DAB = 180^\circ - x^\circ \text{ (opp } \angle \text{'s in cyclic quad } ABCO \text{ are supplementary)}$$

$$\angle EAC = 180^\circ - x^\circ \text{ (} \angle \text{'s at circumference standing on arc CE)}$$

$$\text{Since } \angle DAB = \angle EAC = 180^\circ - x^\circ \text{ then}$$

$$\angle CAB = \angle OAC \text{ (as } \angle DAC \text{ is common)}$$

$$6b) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

Grad of tangent at $(2ap, ap^2)$ is

$$\frac{dy}{dx} = p$$

$\therefore$ Grad of normal is $-\frac{1}{p}$

Equation of normal:

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

✓

(ii) When $x=0$

$$y = \frac{2ap + ap^3}{p}$$

$$y = 2a + ap^2$$

$$\therefore Q \text{ is } (0, 2a + ap^2)$$

✓

(iii) P Q

$$(2ap, ap^2) (0, a(2+p^2))$$

$$-2 : 1$$

$$x = \frac{2ap + 0}{-1} \quad y = \frac{ap^2 - 2a(2+p^2)}{-1}$$

$$x = -2ap \quad y = \frac{-4a - ap^2}{-1} \quad \textcircled{1} x$$

$$x = -2ap, y = 4a + ap^2 \quad \textcircled{1} y$$

$$\therefore R \text{ is } (-2ap, a(4+p^2)) \quad \text{or internal division } \textcircled{1}$$

(iv) $x = -2ap \rightarrow p = \frac{x}{-2a}$ ✓ attempt to

sub in $y = a(4+p^2)$ eliminate parameter

$$y = a\left(4 + \frac{x^2}{4a^2}\right)$$

$$y = 4a + \frac{x^2}{4a}$$

$$4ay = 16a^2 + x^2$$

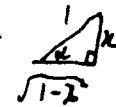
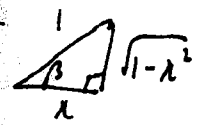
$$x^2 = 4a(y - 4a)$$

✓

Locus is a parabola with vertex ✓

$(0, 4a)$, (focus at $0, 5a$ and

directrix at $y = 3a$)

7a) i) let $\alpha = \sin^{-1} x$ $\sin \alpha = x$ 
let $\beta = \cos^{-1} x$ $\cos \beta = x$ 

$$\text{NB } \alpha, \beta \leq 1$$

$$\sin(\sin^{-1} x - \cos^{-1} x)$$

$$= \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2}$$

$$= x^2 - (1-x^2)$$

$$= 2x^2 - 1$$

$$\text{ii) } \sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$$

$$\sin^{-1} x - \cos^{-1} x = \sin^{-1}(2x^2 - 1)$$

$$\text{then if } \sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$$

$$2x^2 - 1 = 1 - x$$

$$2x^2 + x - 2 = 0$$

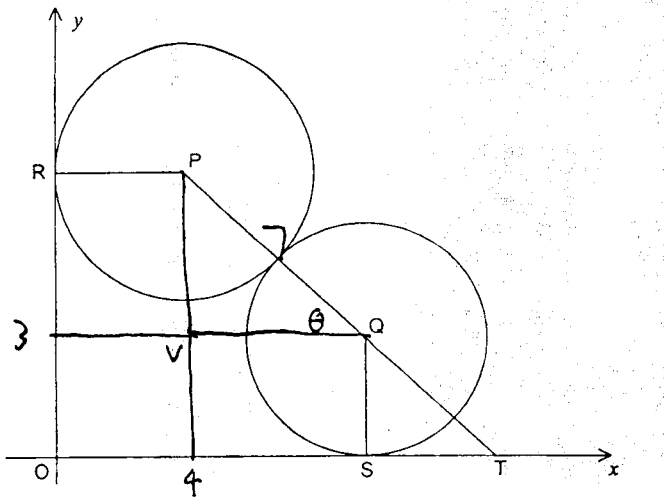
$$x = \frac{-1 \pm \sqrt{1+16}}{4}$$

$$x = \frac{-1 \pm \sqrt{17}}{4}$$

$$\therefore x = \frac{-1 + \sqrt{17}}{4} \quad \text{as } 0 \leq x \leq 1 \quad \checkmark$$

7b) next page

7b)



$$i) OR = 3 + PV \\ = 3 + 7 \sin \theta \quad \checkmark$$

$$OS = 4 + VQ \\ = 4 + 7 \cos \theta \quad \checkmark$$

$$ii) RS^2 = OR^2 + OS^2 \\ = (3 + 7 \sin \theta)^2 + (4 + 7 \cos \theta)^2 \quad \checkmark \\ = 9 + 42 \sin \theta + 49 \sin^2 \theta + 16 + 56 \cos \theta + 49 \cos^2 \theta \\ = 25 + 49(\sin^2 \theta + \cos^2 \theta) + 42 \sin \theta + 56 \cos \theta \quad \textcircled{1} \\ = 74 + 42 \sin \theta + 56 \cos \theta \quad \checkmark$$

$$iii) r \cos(\theta - \alpha) = 42 \sin \theta + 56 \cos \theta \\ r = \sqrt{42^2 + 56^2} \\ r = 70 \quad \checkmark$$

$$\cos \theta \cos \alpha + \sin \theta \sin \alpha = \frac{42}{70} \sin \theta + \frac{56}{70} \cos \theta$$

$$\therefore \sin \alpha = \frac{42}{70} \quad \cos \alpha = \frac{56}{70}$$

$$\tan \alpha = \frac{3}{4} \quad \checkmark$$

$$\alpha = 36^\circ 52' \text{ (nearest min)}$$

$$\therefore RS^2 = 74 + 70 \cos(\theta - 36^\circ 52')$$

N) max value occurs when

$$\cos(\theta - 36^\circ 52') = 1$$

$$\therefore RS^2 = 74 + 70 \\ = 144$$

$$\therefore RS = 12 \text{ gives max value. } \checkmark$$

$$\therefore 12^2 = 74 + 70 \cos(\theta - 36^\circ 52')$$

$$\theta = 36^\circ 52' \quad \textcircled{1}$$